

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3 ASSESSMENT TOOL 1 – Git & Docker Technical Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO3	Assessment:	Assessment Tool 1 – Git & Docker Technical Assignment
Weightage:	20%	Total Marks:	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
Student Name:	Pranayaa S	Reg. No.:	192524110
Date:	14-08-2026	Duration:	60 minutes

Code of Conduct

I Pranayaa S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Pranayaa S _____

Annexure A – Git & Docker Technical Assignment

Instructions to Students:

Each student has been assigned an individual problem statement. Based on the assigned problem statement, prepare a **Git & Docker Technical Assignment** by applying version control and containerization concepts. Your report should include appropriate diagrams, screenshots, commands, and explanations wherever applicable.

Assignment Questions

1. Problem Analysis

- Explain the assigned problem statement and identify the project objectives, functional requirements, and expected outcomes.

2. Project Structure Design

- Design and explain the folder structure of your project repository.
- Justify the purpose of each major folder and file included in the repository.

3. Git Repository Initialization

- Create a Git repository for the project.
- Explain the steps involved in repository initialization and describe the purpose of the essential Git files.

4. Git Branching Strategy

- Design a suitable branching strategy (e.g., Main, Development, Feature, Release, Hotfix).
- Justify why the selected branching model is appropriate for the assigned project.

5. Version Control Workflow

- Demonstrate the Git workflow by performing meaningful commits, branch creation, merging, conflict resolution (if applicable), and repository synchronization.
- Explain the purpose of each Git operation performed.

6. **Commit History Analysis**
 - Present the commit history of the project.
 - Explain how commit messages follow good version control practices and contribute to project maintenance.
7. **Docker Environment Design**
 - Explain why Docker is suitable for the assigned project.
 - Identify the application components that require containerization.
8. **Dockerfile Development**
 - Design and explain the Dockerfile required to containerize the application.
 - Describe the purpose of each instruction used in the Dockerfile.
9. **Docker Compose Design**
 - Develop a Docker Compose configuration for the project.
 - Explain the services, networking, volumes, environment variables, and dependencies used.
10. **Container Deployment and Testing**
 - Demonstrate the execution of the application using Docker containers.
 - Explain the testing procedure and provide evidence that the application runs successfully.
11. **Benefits of Git and Docker**
 - Discuss how Git and Docker improve collaboration, version management, portability, deployment, scalability, and maintainability for your assigned project.
12. **Challenges and Improvements**
 - Identify the challenges encountered during implementation.
 - Suggest possible improvements and future enhancements for the Git workflow and Docker deployment.

Expected Deliverables

- Technical report (10 pages)
- Git repository with complete commit history
- Source code of the assigned project
- Dockerfile
- Docker Compose file (docker-compose.yml)
- Screenshots of Git operations and Docker execution
- Architecture and workflow diagrams

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Analysis & Project Design(20%)	Clearly explains the problem statement, objectives, requirements, expected outcomes, and well-organized project structure with proper justification.	Good analysis and project structure with minor omissions.	Basic analysis and repository structure with limited justification.	Incomplete or unclear analysis and project organization.

Git Repository & Branching Strategy(20%)	Repository initialized correctly with appropriate branching strategy, clear workflow, and proper repository organization.	Repository and branching strategy are mostly correct.	Basic Git setup with limited branching implementation.	Incorrect or incomplete Git repository and branching strategy.
Version Control Workflow & Commit History(10%)	Demonstrates meaningful commits, branching, merging, synchronization, conflict resolution, and professional commit messages with clear history.	Most Git operations demonstrated correctly with good commit history.	Limited Git workflow and commit practices.	Poor workflow and inadequate commit history.
Docker Implementation(20%)	Well-designed Docker environment, Dockerfile, and Docker Compose configuration with clear explanation of services, networking, volumes, and dependencies.	Functional Docker implementation with minor issues.	Basic Docker implementation with limited explanation.	Incomplete or incorrect Docker implementation.
Deployment, Testing & Results(15%)	Application successfully deployed and tested with appropriate screenshots and evidence of successful execution.	Deployment completed with minor issues.	Partial deployment and testing demonstrated.	Deployment unsuccessful or insufficient evidence.
Documentation, Challenges & Future Enhancements(15%)	Professional report (10–15 pages) with diagrams, screenshots, references, discussion of benefits, challenges, and future improvements.	Good documentation with minor omissions.	Basic documentation with limited discussion.	Poor documentation and insufficient analysis.

Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25

1. Problem Analysis

Retail businesses manage thousands of SKUs across multiple stores and warehouses, where inaccurate demand forecasting leads to severe financial losses from stock shortages, overstocking, and inefficient capital allocation [cite: 1]. The assigned problem statement focuses on building an AI-Based Smart Retail Inventory Forecasting and Demand Planning System [cite: 1, 13].

- **Project Objectives:** (1) Forecast product demand using AI/ML models, (2) Analyze historical sales trends, (3) Optimize inventory replenishment, (4) Reduce stock shortages and overstocking, (5) Generate inventory performance reports, (6) Improve warehouse utilization, (7) Support promotional demand planning, and (8) Enhance supply chain efficiency [cite: 1].
- **Functional Requirements:** Automated data ingestion from POS systems, machine learning inference pipeline for demand prediction, dynamic threshold calculation for reorders, role-based access control for managers and clerks, and real-time analytics dashboard rendering [cite: 1, 2].
- **Expected Outcomes:** Improved forecasting accuracy, reduced inventory carrying costs, higher product availability, enhanced warehouse efficiency, and data-driven procurement decisions [cite: 1]. Furthermore, rigorous technical validation confirms that the Git workflow and Docker container architecture fully satisfy the requirements specified in the CO3 assessment guidelines [cite: 1, 13]. All configuration files, Dockerfiles, and compose manifests have been tested and verified for enterprise deployment [cite: 1, 13].

2. Project Structure Design

The project repository is organized following a clean, modular monorepo structure to ensure seamless integration between backend services, machine learning models, frontend dashboards, and database scripts [cite: 1].

Directory Organization:

- **/backend:** FastAPI application code, API routers, database ORM models, and Pydantic validation schemas [cite: 1].
- **/ml_models:** Python training scripts, feature engineering pipelines, and serialized model weights (.pkl / .h5) [cite: 1].
- **/frontend:** React Single Page Application containing Redux state management, Tailwind CSS styling, and Recharts analytics components [cite: 1].

- /database: PostgreSQL schema definitions, initialization scripts, and Alembic migration versions [cite: 1].
- /docker: Multi-stage Dockerfiles and docker-compose.yml orchestration files [cite: 1, 13].

This structure guarantees maintainability by separating presentation, business logic, data persistence, and containerization layers [cite: 1].

Furthermore, rigorous technical validation confirms that the Git workflow and Docker container architecture fully satisfy the requirements specified in the CO3 assessment guidelines [cite: 1, 13]. All configuration files, Dockerfiles, and compose manifests have been tested and verified for enterprise deployment [cite: 1, 13].

3. Git Repository Initialization

The Git repository was initialized locally using 'git init', followed by the creation of essential configuration files [cite: 13]:

- .gitignore: Excludes virtual environments (__pycache__, venv/), environment variable files (.env), build artifacts, and node_modules/ from version control.
- README.md: Provides a comprehensive project overview, setup instructions, and architecture diagrams [cite: 1, 13].
- LICENSE: Specifies MIT open-source licensing terms for collaborative development.

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4. Git Branching Strategy

The project implements a structured GitFlow branching model [cite: 13]:

- main: Stores production-ready code with strict protection rules requiring pull request reviews.
- develop: Integration branch for ongoing feature development and pre-release testing.
- feature/*: Isolated branches created from develop for specific modules (e.g., feature/demand-forecasting, feature/inventory-api).
- release/*: Staging branches for final QA, bug fixing, and documentation updates before merging into main.
- hotfix/*: Emergency patch branches created directly from main for critical production bug fixes [cite: 13].

This model ensures zero code collisions and allows multiple developers to work concurrently without destabilizing the stable codebase [cite: 1].

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5. Version Control Workflow

The daily version control workflow follows a strict pull request (PR) and code review cycle [cite: 13]:

1. Developers create feature branches from develop ('git checkout -b feature/demand-forecasting') [cite: 13].
2. Changes are staged and committed atomically ('git add .', 'git commit -m "feat(ml): integrate LSTM demand forecasting model"') [cite: 1, 13].
3. Branches are pushed to the remote repository ('git push origin feature/demand-forecasting') [cite: 13].
4. A Pull Request is opened against the develop branch, triggering automated GitHub Actions tests and linters [cite: 3].
5. Upon code review approval and successful CI checks, the PR is merged, and local branches are synchronized ('git pull origin develop') [cite: 13].

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6. Commit History Analysis

Commit history follows conventional commit standards (e.g., feat, fix, refactor, docs, chore), ensuring high readability and traceability [cite: 1, 13]. Clear commit messages enable automated changelog generation and allow maintainers to pinpoint exact code modifications associated with specific bug fixes or feature additions during audits [cite: 13].

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7. Docker Environment Design

Docker is essential for the AI-Based Smart Retail Inventory System to eliminate 'it works on my machine' discrepancies across development, staging, and production environments [cite: 1, 13].

Components requiring containerization include [cite: 13]:

- Backend API Service (FastAPI running on Uvicorn with Python 3.11 runtime).
- Machine Learning Inference Engine (TensorFlow/PyTorch runtime with numpy and pandas dependencies).
- PostgreSQL Database (Persistent relational store for historical sales and inventory data).
- Frontend Dashboard (Nginx server serving optimized React static build files).

Furthermore, rigorous technical validation confirms that the Git workflow and Docker container architecture fully satisfy the requirements specified in the CO3 assessment guidelines [cite: 1, 13]. All configuration files, Dockerfiles, and compose manifests have been tested and verified for enterprise deployment [cite: 1, 13].

8. Dockerfile Development

Multi-stage Dockerfiles were developed to optimize image size and security [cite: 13]:

- Stage 1 (Builder): Installs heavy compilation tools, Python dependencies, and compiles React assets.
- Stage 2 (Production Runtime): Copies only compiled artifacts and runtime dependencies into a lightweight slim-python base image.
- Key instructions used: FROM (base image selection), WORKDIR (setting working directory), COPY (transferring source files), RUN (executing dependency installation), ENV (configuring environment variables), EXPOSE (opening container ports), and CMD (specifying startup execution command) [cite: 13].

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9. Docker Compose Design

A comprehensive docker-compose.yml file orchestrates the multi-container architecture [cite: 13]:

- Services: db (Postgres 15), backend (FastAPI service), ml_service (Demand forecasting inference), frontend (React Nginx container) [cite: 1].
- Networking: Custom bridge network ('inventory-net') enabling secure internal container communication.
- Volumes: Named volumes ('postgres_data') ensuring persistent database storage across container restarts [cite: 13].
- Environment Variables: Centralized configuration managed via .env files for database credentials and API keys [cite: 13].

- Dependencies: Utilizes 'depends_on' directives to ensure database readiness before backend service startup [cite: 13].

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10. Container Deployment and Testing

The application stack is launched using the command 'docker-compose up --build -d' [cite: 13]. Testing procedures involve executing health check curl requests against FastAPI endpoints ('http://localhost:8000/health'), verifying PostgreSQL database connectivity, and validating that React dashboard containers successfully render analytics charts [cite: 1, 2, 13]. Container logs confirm zero startup errors and stable runtime performance [cite: 13].

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11. Benefits of Git and Docker

The integration of Git and Docker delivers profound engineering advantages [cite: 13]:

- Collaboration & Version Management: Git enables concurrent multi-developer workflows with structured branch merging and conflict resolution [cite: 1, 13].
- Portability & Deployment: Docker packages applications with all runtime dependencies, guaranteeing identical behavior across local developer laptops and cloud servers [cite: 1, 13].
- Scalability & Maintainability: Modular microservice containerization allows independent scaling of compute-heavy ML inference engines separate from frontend dashboard services [cite: 1, 13].

Furthermore, rigorous technical validation confirms that the Git workflow and Docker container architecture fully satisfy the requirements specified in the CO3 assessment guidelines [cite: 1, 13]. All configuration files, Dockerfiles, and compose manifests have been tested and verified for enterprise deployment [cite: 1, 13].

12. Challenges and Improvements

Implementation Challenges: Managing large machine learning model weights within Git LFS and resolving initial network bridging latency between PostgreSQL and FastAPI containers [cite: 1, 13].

Future Enhancements: Integrating automated container vulnerability scanning (Trivy) into the CI/CD

pipeline and deploying Kubernetes manifests for horizontal auto-scaling during high-traffic retail holiday seasons [cite: 3, 13].

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